Machine Learning vs. Deep Learning: Two Real-World Cases

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Machine learning (ML) and deep learning (DL) both learn from data, but they differ in who finds the features. In traditional ML, a person chooses the features and an algorithm such as linear regression learns from them. In DL, a multi-layer neural network learns the features directly from raw data. As the lesson puts it, ML is like giving a student a checklist, while DL is like teaching the student to read and letting them learn from experience.

This report compares two cases from the lesson. Customer churn prediction with a support vector machine (SVM) is the machine learning example. Image recognition with a convolutional neural network is the deep learning example, applied to diabetic eye disease screening. For each case, I explain why the chosen approach fits the problem and why the other approach does not.

Example 1 (Machine Learning): Customer Churn Prediction

Lesson Example

A support vector machine (SVM) predicts whether a customer is likely to leave a service. It learns from historical records using features a person chooses, such as customer age, contract length, and monthly charges.

Real-World Application

Subscription businesses such as phone carriers, streaming services, and newspapers lose revenue when customers cancel, which is called churn. Coussement and Van den Poel (2008) built an SVM churn model for a newspaper subscription service using subscription details and customer-interaction records. The SVM performed well on noisy marketing data and, once its parameters were tuned, beat logistic regression; a random forest, another traditional ML method, did even better. The study also identified the main drivers of churn. A company can use such a model to list high-risk customers and send them retention offers before they leave.

Why Machine Learning Fits This Problem

- **The data is structured.** Each customer is a row in a table with clear columns such as contract length, monthly charges, and number of complaints. Domain experts already know which features matter, so manual feature selection works well.
- **The results must be explained.** Managers need to know why customers leave so they can fix the cause, such as high prices or short contracts. Traditional ML models show which features drive the prediction.
- **The data size is moderate.** A business may have thousands or tens of thousands of customer records, which is enough for an SVM.
- **Cost is low.** The model trains quickly on an ordinary computer and can be retrained every month as customer behavior changes.

Why Deep Learning Does Not Fit

Deep learning's main strength is learning features from raw images, audio, or text, but churn data is already a clean table of known features. On medium-sized tables like this, traditional ML models often match or beat deep learning and train much faster. A deep network would also need more data and GPU power, and its millions of weights would not tell managers why a customer is likely to leave.

Example 2 (Deep Learning): Image Recognition

Lesson Example

A convolutional neural network (CNN) learns to recognize objects in images directly from raw pixels. Its early layers detect edges and textures, and its deeper layers combine them into complex shapes, with no hand-defined features.

Real-World Application

Diabetic retinopathy is diabetes-related damage to the retina and a leading cause of vision loss. In 2016, Google researchers published a study in *JAMA* in which a deep CNN was trained on 128,175 retinal photographs, each graded by eye doctors. The model detected referable diabetic retinopathy with accuracy comparable to board-certified ophthalmologists (Gulshan et al., 2016). In April 2018, the U.S. Food and Drug Administration (FDA) authorized IDx-DR, the

first AI device that gives a retinopathy screening result without a specialist reading the image (U.S. Food and Drug Administration, 2018). This lets a primary care office screen patients and refer only those who need an eye specialist.

Why Deep Learning Fits This Problem

- **The input is raw images.** Signs of disease include tiny hemorrhages, swelling, and abnormal vessels that vary in size, shape, and position. A CNN learns these visual patterns from pixels by itself.
- **A large labeled dataset exists.** Diabetes screening programs produce many graded photographs, which is the amount of data deep learning needs.
- **Accuracy matters most.** Missing disease can mean blindness, and deep learning reached specialist-level sensitivity. Specialized hardware (GPUs) makes the heavy training practical.

Why Traditional Machine Learning Does Not Fit

A traditional model such as an SVM or decision tree needs humans to turn each image into features first, for example by counting red spots or measuring vessel width. Writing reliable rules for every lesion type, camera, and lighting condition is slow and fragile. Hand-built features also miss subtle patterns that experts cannot easily describe. Gulshan et al. (2016) note that deep learning avoids this manual feature engineering by learning the most useful features straight from labeled images, so traditional ML is the weaker choice here.

Comparison and Conclusion

The right approach depends on the data and on what the result must do. Table 1 applies the lesson’s three key differences to both cases.

Table 1

Key Differences Between Machine Learning and Deep Learning in the Two Cases

Key difference	Customer churn prediction (ML)	Retinopathy screening (DL)
Feature extraction	Experts choose known features (contract length, monthly charges, age)	The CNN learns features from raw pixels
Data requirements	Thousands of customer records; runs on a normal computer	More than 128,000 labeled images; needs GPUs
Complexity	Simpler model; managers can see which factors drive churn	Millions of parameters; high accuracy but hard to explain

Machine learning is the better choice when data is structured, moderate in size, and the decision must be explained. Deep learning is the better choice when the input is raw, unstructured data such as images, large labeled datasets exist, and accuracy matters more than explaining each prediction.

References

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